Learning the Rhythm: Why Rotary Position Encoding Degrades Beyond the Training Horizon, and How Learnable Frequencies Help

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Abstract

Forecasters in finance and energy need predictions at several horizons at once, yet a model is usually trained for one horizon. A Transformer that represents each future step by a placeholder token, distinguished only by its rotary position encoding (RoPE), can in principle forecast any horizon. In our experiments with standard RoPE it does not: trained for 96 steps and asked for 720 (on hourly data, 4 versus 30 days ahead), its error reaches up to 6.4 times that of a model trained for 720. Our proposal's learnable time scale per attention head (4 parameters per layer) reduces this error by 38–66% on three seasonal benchmarks. The evidence indicates that the learned scales move RoPE's fastest rotation close to the daily cycle, so distant positions resemble positions seen in training. On ETTm1, fixing the scale at the daily-aligned value without learning gives the same gain, while values 5% off do not. The gain is consistent across seeds and significant by Diebold–Mariano tests on strongly seasonal data, including two electricity markets; on exchange rates, which have no daily cycle, we find no consistent effect. A single model trained on mixed horizons comes within 3% of models trained per horizon.

1 Problem and how we arrived at it

Risk managers report Value-at-Risk at 1 and 10 days; energy traders bid for the next day and hedge the next week. Maintaining one model per horizon is costly, so a model that is trained once and forecasts any horizon is useful — for example, trained to forecast hourly electricity prices 4 days ahead and then asked for 30 days. Our proposal asked whether a Transformer could *learn* the time scales that matter, by giving each attention head a learnable multiplier on its rotary position frequencies. In our experiments the evidence for that hypothesis was weak: the learned scales stayed within ±10% of 1, heads learned similar scales, and accuracy at the training horizon changed little. The same scales, however, made a large difference when the model forecast *beyond* its training horizon.

2 Model

Backbone. Each univariate history of L=336 steps is cut into non-overlapping patches of P=16 steps, projected to d=64 dimensions, and processed by a 2-layer, 4-head pre-norm Transformer encoder. After the 21 history tokens we append one learnable placeholder token per future patch (6 for H=96, 45 for H=720); all placeholders share the same vector, so *only their position encodings tell them apart*. History tokens attend to history only; each placeholder attends to all history and to itself, so the forecast for step h does not depend on the total horizon and the model can be queried at any H after training. A shared linear layer reads each placeholder out into P forecast values. We use patches because point tokens gave noisier extrapolation (Sec. 6).

Rotary position encoding. RoPE [1] rotates each pair j of query/key coordinates at token position t by the angle $t\,\omega_j$ with $\omega_j = b^{-j/m}$, $m = d_h/2 = 8$, base $b \in \{10^3, 10^4\}$. Because $\langle R(t\omega)q, R(u\omega)k\rangle$ depends only on $t - u$, attention sees relative distances. The fastest pair rotates at $\omega_0 = 1$ rad/token, i.e. it repeats every $2\pi \approx 6.28$ tokens.

Learnable frequency scale (main method). Each layer ℓ and head h has a scalar $a_{\ell,h}$, initialised to 0, and uses $\hat{\omega}_{\ell,h,j} = \omega_j \, e^{a_{\ell,h}}$; this adds 4 parameters per layer and starts exactly at standard RoPE. *Variants* used as ablations: *grouped* (separate scales for the 4 fastest and 4 slowest pairs), *shared* (one scale per layer for all heads), and *fixed* scales set by hand and not trained.

Why alignment should matter. If a series repeats every T tokens and $\hat{\omega}_0 = 2\pi/T$, then position $t + T$ has exactly the same fastest-pair angle as t: a placeholder far beyond the training horizon is, to that pair, indistinguishable from one at the same time of day inside it. With the default $\omega_0 = 1$ and, e.g., ETTm1 ($T = 6$ tokens = 1 day), the angle drifts by 0.28 tokens per day and the model meets angle combinations it never saw.

Multi-horizon training. As a second remedy, each training batch uses a horizon drawn from $\{96, 192, 336, 720\}$, so one model sees distant placeholders during training.

3 Data, protocol and baselines

Data (sources and URLs in Appendix B). Four standard benchmarks — Electricity (321 clients, hourly), ETTh1/ETTm1 (transformer load, hourly/15-min), Weather (21 variables, 10-min) — and three financial markets: day-ahead electricity prices of Nord Pool (EPF-NP) and PJM (EPF-PJM), hourly 2013–2018, and daily exchange rates of 8 currencies. A *step* is one observation. Following the long-term forecasting benchmarks [2, 3], models read L=336 past steps and forecast $H \in \{96, 192, 336, 720\}$ steps; we train at H=96 and test up to 720. In calendar time, 96→720 steps is 4→30 days on hourly data (Electricity, ETTh1, EPF), 1→7.5 days on ETTm1 and 16 h→5 days on Weather; on daily exchange rates 720 steps (≈3 years) exceeds any practical horizon and is kept only for a uniform protocol. Every variable is forecast as its own series. Splits are chronological (ETT: 12/4/4 months; others 70/10/20%); z-scores use training rows only; for exchange rates we add per-window normalisation (RevIN), without which every model fails on this drifting series. We report MSE on the normalised scale.

Protocol. All models share the backbone, optimiser (AdamW, lr 10^{-3}, warm-up + cosine), batch order, initial weights (paired seeds) and a fixed step budget per dataset. The RoPE base and the best checkpoint are chosen on the validation set only; the test set is used once. We use 3 seeds (5 on ETTh1 ablations).

Baselines and why. (i) *Fixed RoPE*, the same model without the scales, isolates the mechanism; (ii) a channel-wise *linear* model $\mathbb{R}^{336} \to \mathbb{R}^{H}$, because simple linear models match Transformers on these benchmarks [2]; (iii) models *trained directly* at H=720, the natural alternative to extrapolation.

Significance. We report how many of the 6 runs (3 seeds × 2 bases) improve, since a test within one run ignores variation between runs, and a Diebold–Mariano test [4] per run on the per-origin loss difference (steps 97–720; Newey–West variance, lag 719).

4 Results

4.1 At the training horizon the model is competitive, and the scales change little

The Transformer is close to published results; it beats the linear model on Weather and EPF-NP, ties on Electricity, and loses on ETTh1, ETTm1 and exchange rates, in line with [2]. At H=96, fixed and learnable RoPE differ by about one standard deviation across seeds or less. Weather's lower test than train error reflects a calmer test period; splits are strictly chronological.

Table 1: Train and test MSE at the training horizon H=96 (mean±std over seeds). For reference, PatchTST/42 (L=336) reports Electricity 0.130, ETTh1 0.375, ETTm1 0.290, Weather 0.152 ([3], Table 3).

	Linear		Transformer, fixed RoPE		Transformer, learnable scale	
Dataset	train	test	train	test	train	test
Electricity	0.139±0.000	0.140±0.000	0.133±0.001	0.139±0.001	0.133±0.001	0.139±0.001
ETTh1	0.336±0.001	0.377±0.002	0.338±0.013	0.417±0.017	0.340±0.010	0.414±0.011
ETTm1	0.272±0.003	0.304±0.004	0.223±0.010	0.311±0.009	0.219±0.008	0.317±0.011
Weather	0.454±0.001	0.174±0.000	0.388±0.006	0.153±0.000	0.388±0.004	0.153±0.001
EPF-NP	0.222±0.000	0.472±0.000	0.149±0.014	0.482±0.032	0.139±0.006	0.452±0.013
EPF-PJM	0.637±0.001	0.163±0.000	0.235±0.008	0.174±0.015	0.378±0.143	0.167±0.005
Exchange	0.122±0.000	0.089±0.000	0.120±0.022	0.103±0.010	0.114±0.022	0.099±0.005

4.2 Beyond the training horizon, fixed RoPE degrades and learnable scales reduce it

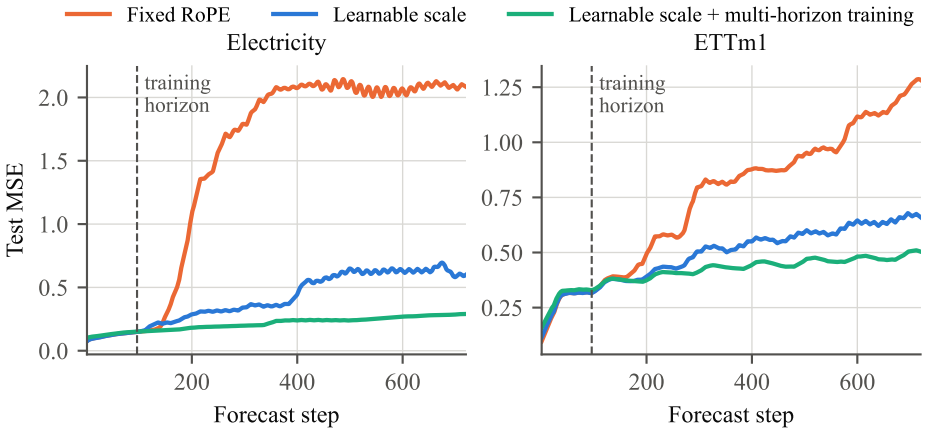


Figure 1: Test MSE per forecast step for models trained at H=96 (base 10^4, mean of 3 seeds). Fixed RoPE degrades soon after it leaves the training horizon.

Figure 1 and Table 2 show the central finding. Queried at 720 steps, fixed RoPE reaches 1.5–6.4 times the error of a Transformer trained at 720 on Electricity, ETTh1 and ETTm1; the learnable scale reduces its error by 38–66%, with improvement in all 6 runs and a significant DM test in each. On ETTh1 the extrapolating model also has lower error than both models trained directly at 720.

4.3 Why: evidence that the learned scales align RoPE with the daily cycle

Four observations support one explanation. **(1) Learning matters more than head diversity.** For extrapolation on ETTh1, one scale shared by all heads performs as well as one per head, and hand-set spreads of head scales help little or not at all (Table 3); we found no evidence that the gain requires heads to specialise to time scales. **(2) The effect sits in the fast frequencies.** In the grouped variant the fast group learns almost the same value as the per-head scale, while the slow group drifts (0.95–1.13) with no measurable effect on the forecast. **(3) The learned value is close to the aligning value.** On ETTm1 one day is 6 tokens; the learned scales average 1.04 against the aligning $2\pi/6 = 1.047$. A fixed scale of 1.047 gives the same gain without learning, while 0.955 or 1.10 do not (Fig. 2; 1.047 beats 0.955 in all 6 runs, DM $p < 10^{-10}$). During training the 24 per-head scales move from 1 towards this value (mean 1.036, Fig. 3a), and resetting them to 1 at test time, with all other weights unchanged, brings back the failure of fixed RoPE (Fig. 3b): the gain lives in the scales. **(4) Gains are larger on more seasonal series** (Fig. 4). Across channels the ratio fixed/learnable rises with daily seasonal strength [5]: on ETTm1 the ratio reaches 1.8 on the two most seasonal load channels, and in Weather

Table 2: Test MSE over steps 1–720. Columns 2–3: trained at H=96 and queried at 720. DM: of 6 runs (seed×base), how many are significantly better / worse with learnable scales (p<0.05). Columns 5–7: one multi-horizon model, and models trained directly at 720.

	Trained at 96		DM	Multi-horizon	Trained at 720	
Dataset	fixed	learnable	+/−	learnable	Transformer	Linear
Electricity	1.347±0.087	0.464±0.107	6/0	0.216±0.003	0.210±0.002	0.204±0.000
ETTh1	0.972±0.071	0.473±0.013	6/0	0.541±0.014	0.632±0.036	0.493±0.004
ETTm1	0.764±0.068	0.474±0.019	6/0	0.421±0.011	0.435±0.013	0.430±0.003
Weather	0.392±0.009	0.396±0.006	2/1	0.326±0.009	0.336±0.004	0.325±0.001
EPF-NP	1.072±0.061	0.896±0.039	5/0	1.123±0.008	1.196±0.156	0.863±0.003
EPF-PJM	0.443±0.036	0.273±0.034	6/0	0.216±0.002	0.213±0.003	0.221±0.000
Exchange	1.078±0.100	1.024±0.036	3/3	1.074±0.087	1.097±0.214	0.890±0.003

Table 3: Mechanism ablations, all trained at H=96 (test MSE).

Frequency scales	Test MSE
ETTh1, steps 1–336, 5 seeds	
Fixed RoPE (scale 1)	0.676±0.022
Learnable, per head	0.460±0.006
Learnable, grouped	0.460±0.006
Learnable, shared by heads	0.458±0.011
Fixed, mild spread (0.9–1.1)	0.547±0.038
Fixed, wide spread (0.5–2.0)	0.716±0.064
ETTm1, steps 1–720, 3 seeds × 2 bases	
Fixed RoPE (scale 1)	0.767±0.051
Learnable, per head	0.488±0.033
Fixed 1.047 (= 1 day)	0.459±0.013
Fixed 0.955 (misaligned)	0.923±0.065
Fixed 1.10 (misaligned)	0.809±0.044

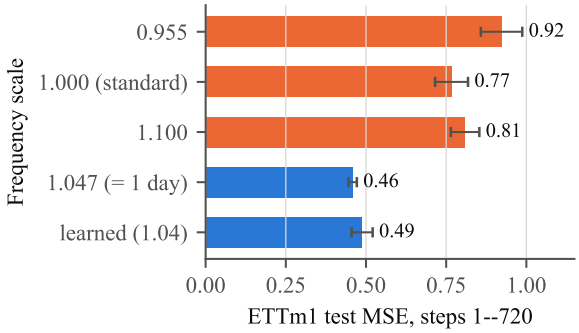


Figure 2: ETTm1, 6 tokens per day: scale $2\pi/6 = 1.047$ aligns RoPE's fastest rotation with the day. Fixed at 1.047 without learning it matches the learned model; 5% too slow or fast collapses like standard RoPE.

only the three solar-radiation channels gain (ratio 1.14–1.17), which explains why Weather as a whole shows no gain. Across datasets the relation is clear at the extremes (Electricity largest, exchange rates none) but loose in between (EPF-PJM gains more than its seasonal strength suggests), so seasonal strength indicates where to expect a gain but does not predict its size. **Attention moves where the RoPE arithmetic predicts** (Fig. 5). With fixed RoPE the fastest pair drifts by $6 - 2\pi \approx -0.28$ rad per day, so beyond the lags seen in training a same-hour patch has the code of a patch 4 h, later 8 h, away. The placeholders' attention follows this model-free prediction: its share on the same hour falls from 0.66 to 0.03 over 7.5 days as it shifts to +4 h and then +8 h, while with 1.047 it stays (0.49) and with the learned scales it decays slowly (0.38). Gradient clipping is not the cause: it is inactive after warm-up.

4.4 How far it goes: multi-horizon training

Learnable scales do not close the whole gap: on Electricity, ETTm1 and Weather the extrapolating model stays above models trained at 720 (Table 2). A single model trained on mixed horizons is within 3% of, or better than, the Transformer trained at 720 on all seven datasets, at a small cost at H=96 (Electricity 0.135 vs 0.128 on the same windows); with it, fixed and learnable RoPE perform alike, presumably because distant positions are seen in training. In our experiments multi-horizon training is the stronger remedy when long training targets are available; learnable scales help when they are not, or when long-horizon training overfits (ETTh1, EPF-NP).

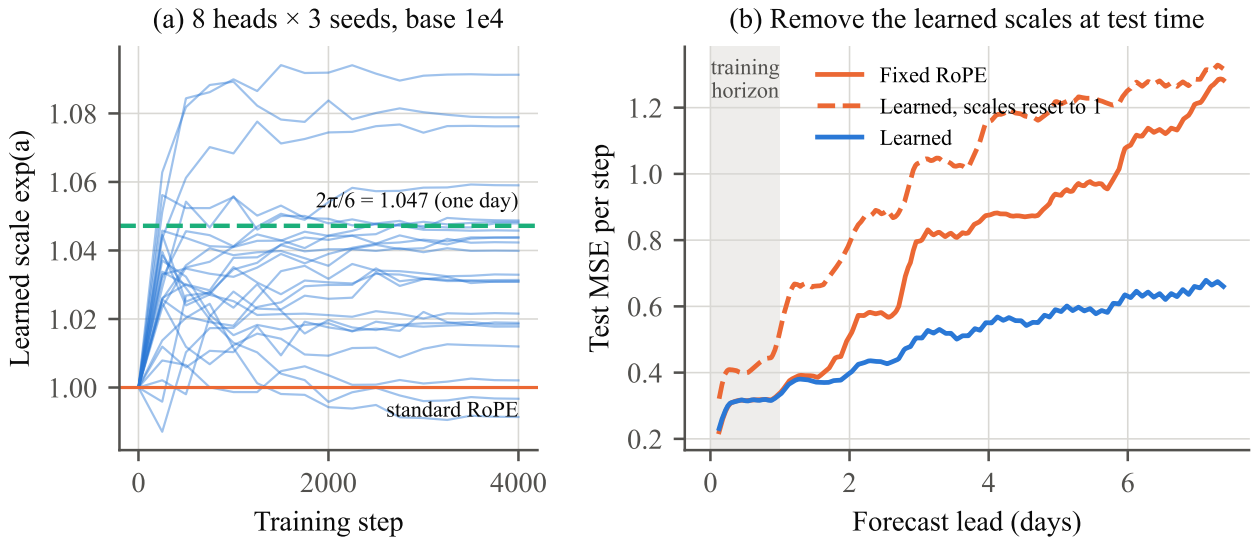


Figure 3: ETTm1, base 10^4. (a) The 24 learned per-head scales (8 heads × 3 seeds) during training; dashed: the day-aligned value $2\pi/6$. (b) Test MSE per step (6-hour moving average) of fixed RoPE, the learnable model, and the same learnable weights with every scale reset to 1 at test time.

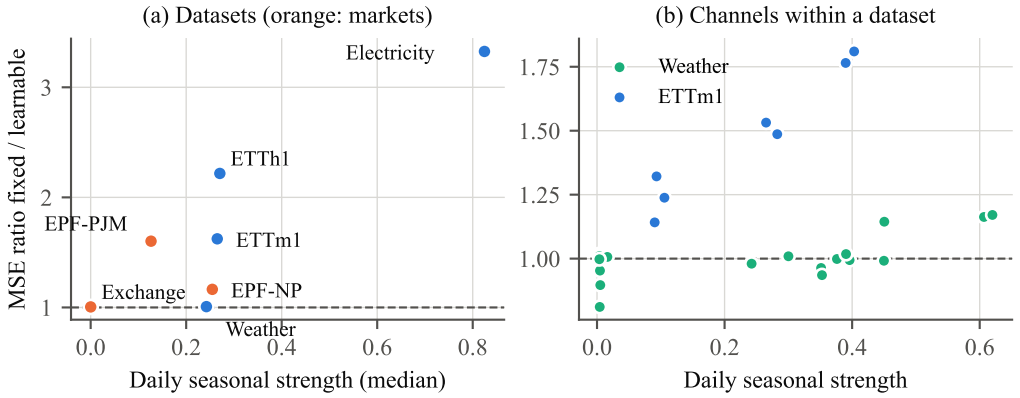


Figure 4: Gain from learnable scales (MSE ratio fixed/learnable on steps 97–720; > 1 means learnable is better) against daily seasonal strength.

5 Application to finance

Our results suggest a practical guideline for financial forecasting: *when extrapolating in horizon, the position encoding should match the calendar cycle of the series, and learnable RoPE scales can learn this match.* We tested this on financial data chosen before seeing any of its results. Electricity prices are traded assets with strong daily cycles; on both markets learnable scales reduce 720-step error (EPF-PJM 0.44→0.27, significant in 6/6 runs; EPF-NP 1.07→0.90, 5/6). Exchange rates behave like a random walk with no cycle; here the scales have no consistent effect (3 runs significantly better, 3 worse), the linear model is best, and RevIN is needed — consistent with the explanation above. In practice: (i) for seasonal series such as intraday volume and volatility or power and gas prices, use learnable or calendar-aligned RoPE before forecasting beyond the training horizon; (ii) to cover several risk horizons (1- and 10-day VaR, daily-to-weekly volatility as in HAR) with one model, train on mixed horizons; (iii) for near-random-walk prices, we saw no benefit from position encodings.

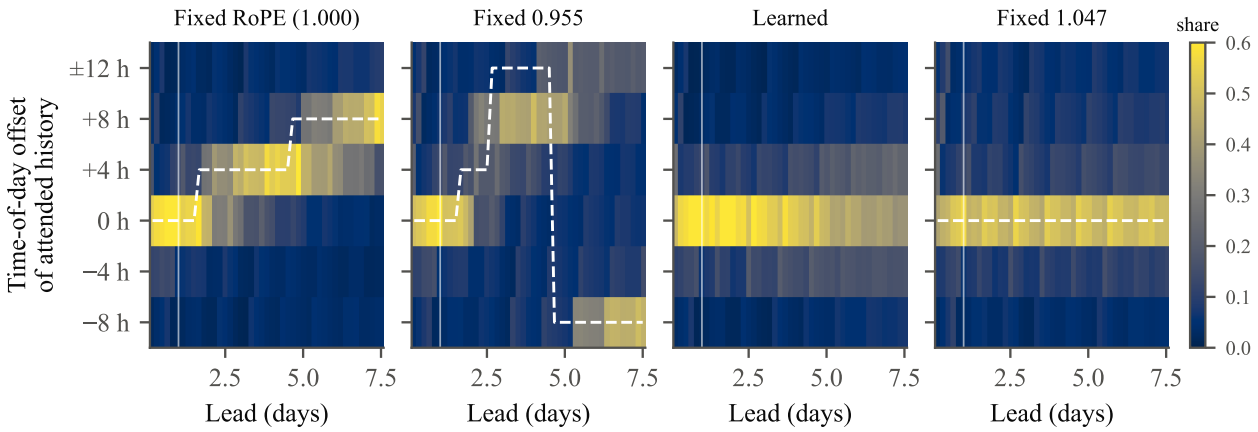


Figure 5: Where the placeholders look (ETTm1, trained at H=96, queried at 720; last layer, mean over heads and 3 seeds). Colour: share of attention on history patches at each time-of-day offset from the forecast step. Dashed: offset predicted from RoPE arithmetic alone — the offset whose fastest-pair angle is closest to a lag seen in training.

6 Limitations and conclusion

We found no evidence that heads specialise to distinct time scales, as the proposal expected, and grouping frequencies gave no extra gain. The model is small, seeds are few, and the mechanism analyses (fixed scales, attention, channels) cover ETTh1, ETTm1 and Weather only; with point tokens the effect is noisier, and linear models stay strong at the training horizon. In short, our evidence indicates that RoPE degrades beyond the training horizon because its rotation is out of step with the data's daily cycle; four learnable parameters per layer largely correct this on seasonal data, and mixed-horizon training narrows the remaining gap further.

References

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Appendix A. Step-by-step replication

All runs used Python 3.13, PyTorch 2.6 (CUDA 12.4) on one NVIDIA H100 (memory capped at 8%); a run takes 20 s–3 min. Code: repository `https://github.com/ycsama0703/ft5011-ropeformer`, branch `final-horizon-extrapolation`, with the scripts in `experiments/`. **Trained models and all result files** (257 checkpoints, 71 MB) are attached to the release `https://github.com/ycsama0703/ft5011-ropeformer/releases/tag/final-report-2026-11-10`; unzipped at the repository root, they reproduce every table and figure with `python report/make_assets.py results` and allow the analyses in step 6 without retraining.

1. `pip install -e .` (PyTorch ≥2.0); `python -m unittest discover -s tests` (29 tests).
2. Data: `python tools/download_data.py ETTh1 ETTm1 weather electricity exchange_rate`; financial data: `python experiments/build_fin.py raw data` after downloading EPF (Appendix B).
3. Train at 96 and extrapolate, with direct-training references, for each dataset D and its step budget S (ETTh1, EPF, exchange: 2000; ETTm1: 4000; Weather: 6000; Electricity: 8000): `python tools/extrap_study.py --dataset D --steps S --patch-len 16 --amp bf16 --seeds 0 1 2 --bases 1000 10000 --modes standard head grouped --eval-horizons 192 336 720 --direct-horizons 336 720 --direct-modes head --out results/r2_D.json` (add `--instance-norm` for exchange rates).
4. Multi-horizon: `python tools/multi_horizon.py --dataset D --steps S --train-horizons 96 192 336 720 --test-horizon 720 --out results/mh_D.json`.
5. Ablations (Table 3): `extrap_study.py` with `--modes shared_group fixed_head:0.9,0.97,1.03,1.1` `fixed_head:0.5,0.8,1.25,2.0` (ETTh1, 5 seeds) and `fixed_head:1.0472,1.0472,1.0472,1.0472` etc. (ETTm1).
6. Train/val/test MSE at 96, linear reference, Diebold–Mariano tests, per-channel analysis and seasonal strength: scripts `train_eval.py`, `linear96.py`, `dm_test.py`, `per_channel.py`, `periodicity.py` in `experiments/` (commands in `experiments/README.md`).
7. All tables and figures of this report: `python report/make_assets.py`; Figures 3 and 5 from the released checkpoints on CPU in about 10 s: `python report/mechanism_figures.py --results results --data-dir data` (the predicted offsets come from `report/phase_figures.py`).

Hyper-parameters: L=336, patch 16, d=64, 4 heads, 2 layers, FFN 128, dropout 0.1, batch 256, AdamW (lr 10^{-3}, weight decay 0.01, none on scales), 300 warm-up steps then cosine decay, gradient clip 1.0, bf16 autocast; validation every 200–500 steps keeps the best state.

Appendix B. Data sources

- ETTh1, ETTm1, Weather, Electricity, exchange rates: Time-Series-Library collection (CC BY 4.0), `https://huggingface.co/datasets/thuml/Time-Series-Library`.
- EPF day-ahead prices (Nord Pool, PJM; hourly 2013–2018) [6]: `https://zenodo.org/records/4624805` (files `NP.csv`, `PJM.csv`); we keep the price column only.
- Bitcoin (supplementary, low seasonality): Binance public 1-hour klines, `https://data.binance.vision`, BTCUSDT 2019–2024; channels log volume, log trade count, |log return|.